

CYCLOPEpDB: Bridging Molecular Dynamics Simulations and Machine Learning for Predicting Cyclic Peptide–Membrane Interactions

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Supplementary Material

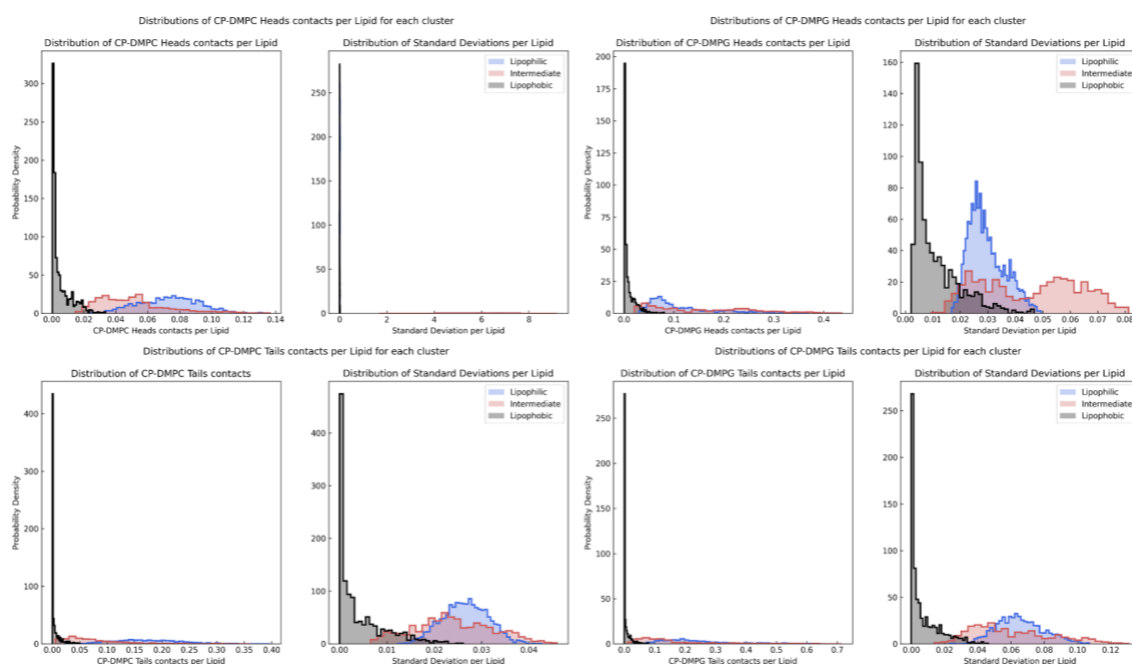


Figure S1.- Distribution of CP-Lipid contacts per lipid type and part, for the clusters obtained from the OPTICS algorithm. The distributions in the left column corresponds to CP-DMPC lipids while the distributions in the right correspond to CP-DMPG lipid contacts. The first column displays distributions for CP-Lipid Heads contacts, whereas the second row shows the distribution for CP-Lipid Tails contacts.

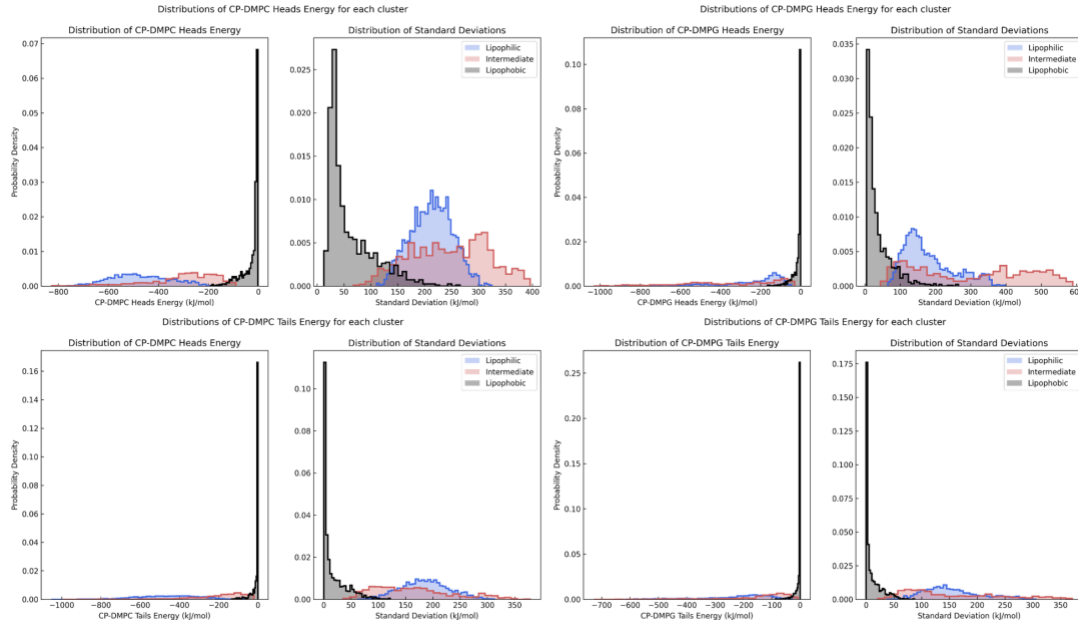


Figure S2.- Distribution of CP-Lipid interaction energy per lipid type and part, for the clusters obtained from the OPTICS algorithm. The distributions in the left column corresponds to CP-DMPC lipids while the distributions in the right correspond to CP-DMPG lipids. The first column displays distributions for CP-Lipid Heads interaction energy, whereas the second row shows the distribution for CP-Lipid Tails interactions.

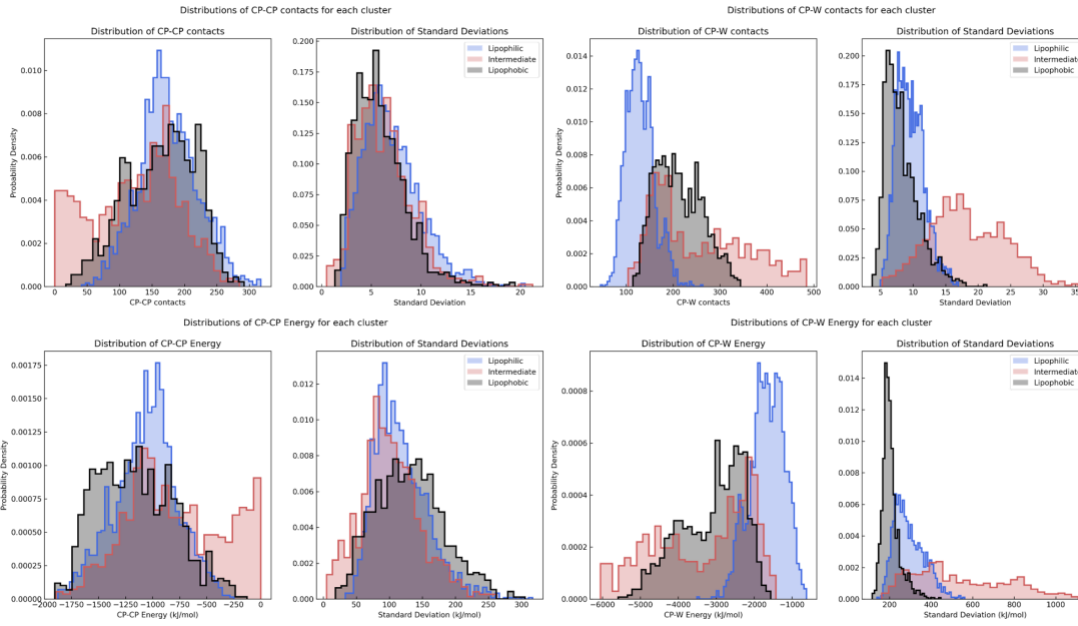


Figure S3.- Distributions of CP-CP and CP-Water for the clusters obtained from the OPTICS algorithm. The distributions in the first row correspond to contacts, while the second row shows the distributions for interaction energies.

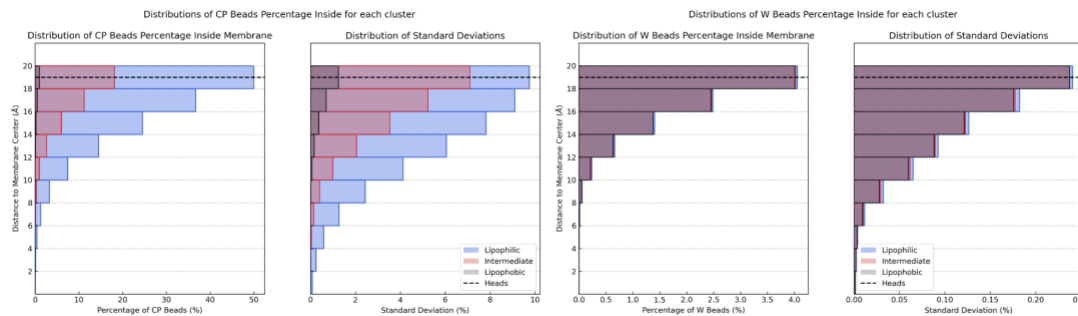


Figure S4.- Distributions of the percentage of CP (left) and Water (right) beads at different distances from the center of mass of the membrane. The horizontal line between 18 and 20 Å indicates the position of the phosphate groups of

the lipids. The number of beads of each group were calculated every 2 Å, from 2 to 20 Å, starting at the lipid bilayer's center of mass, and expressed as the percentage of total beads in the group for fair comparison.

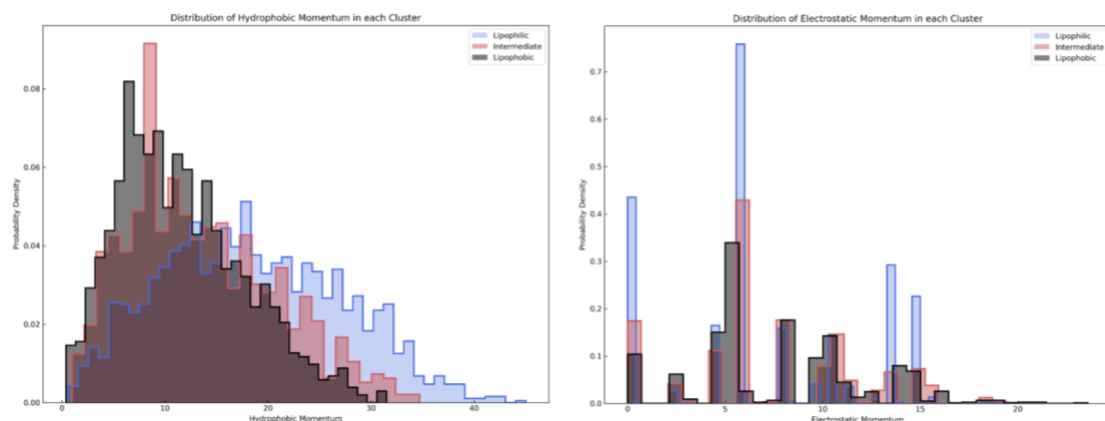


Figure S5.- Distributions of the Hydrophobic momentum (left) and Electrostatic Momentum (right) of the simulated sequences for the clusters obtained from the OPTICS algorithm.

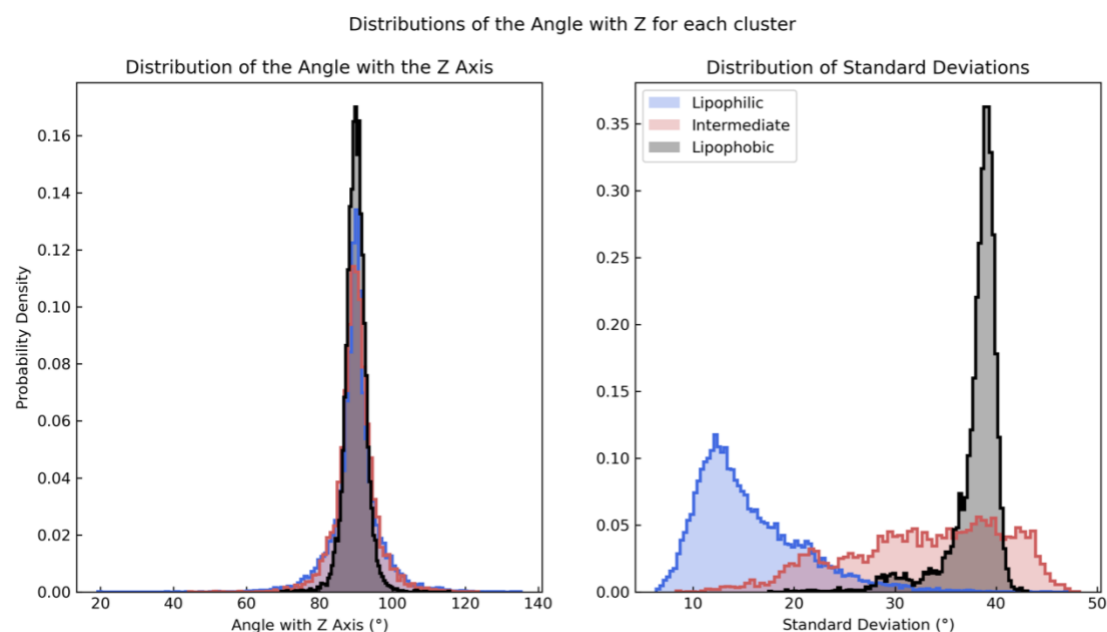


Figure S6.- Distribution of the Angle of the CPs relative to the membrane vertical axis for the clusters obtained from the OPTICS algorithm.

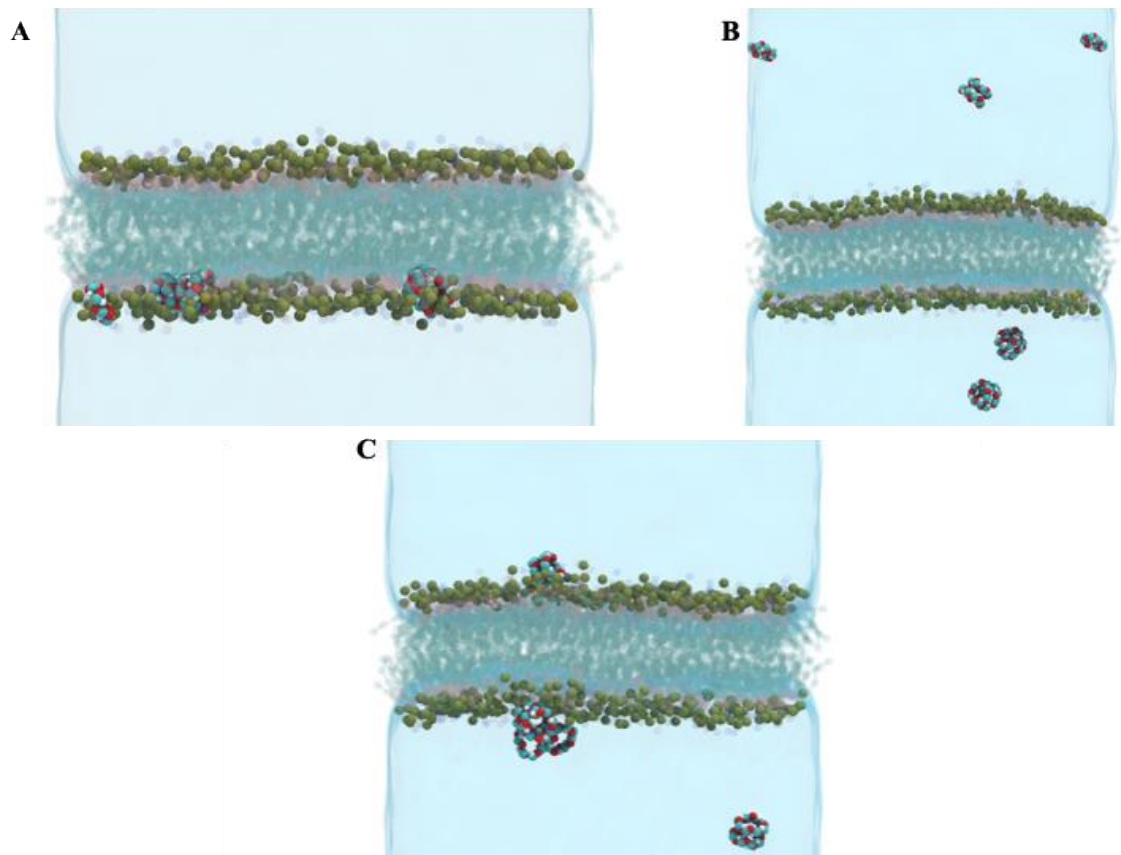


Figure S7.- Snapshots of the last frame (at 10 us) for three random sequences, one for each cluster. All the snapshots correspond to the second replica with an initial rotation of 120 degrees relative to the membrane. Snapshot **A** corresponds to MIIFDACV, classified as Lipophilic, while **B** corresponds to the Lipophobic sequence EEEEYEND, and **C** to KHKNGRRR, from the Intermediate cluster. The backbone of the CP is represented in cyan. The beads colored red and white represent the amino and carbonyl groups of the CPs in the MA(R/S)TINI 3 force field. Phosphate groups, shown in ochre, mark the lipid bilayer's boundaries. The lipid tails are rendered transparently to facilitate visualization, and the water is depicted as a continuous light-blue region.

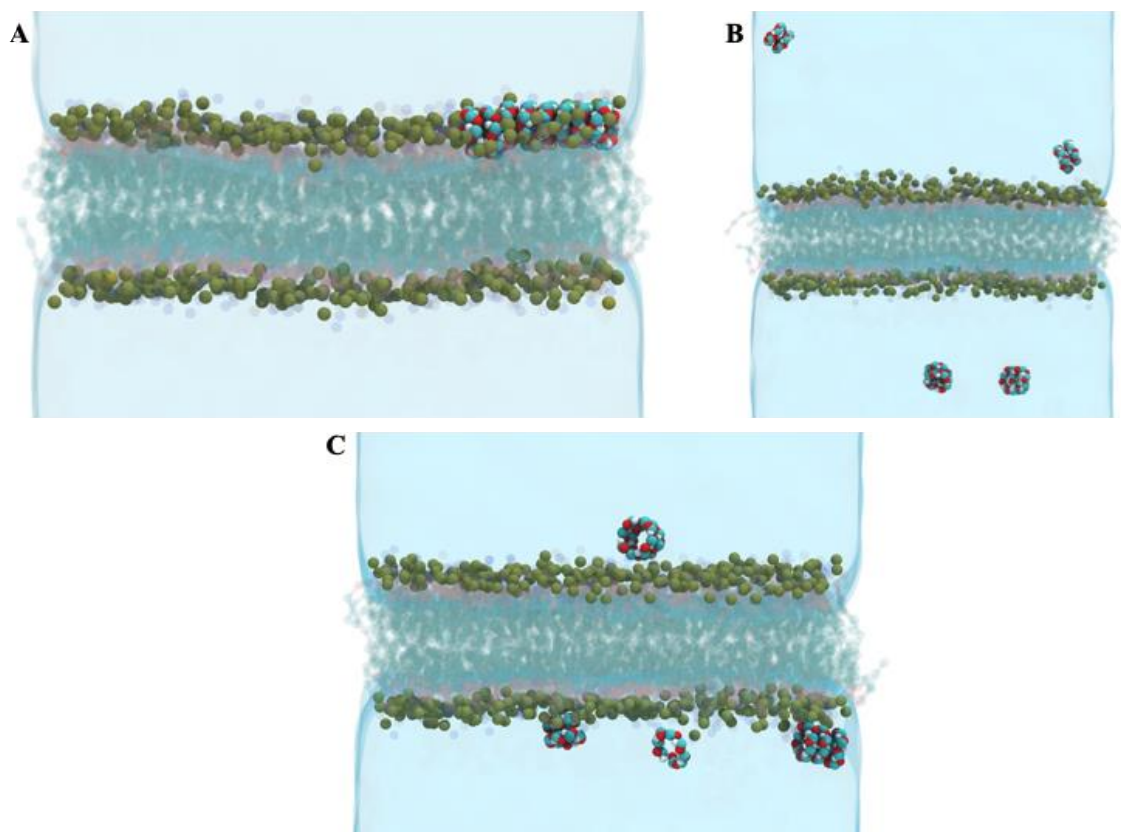


Figure S8.- Snapshots of the last frame (at 10 us) for three random sequences, one for each cluster. All the snapshots correspond to the third replica with an initial rotation of 240 degrees relative to the membrane. Snapshot **A** corresponds

to MIIFDACV, classified as Lipophilic, while **B** corresponds to the Lipophobic sequence EEEEYEEND, and **C** to KHKNGRRR, from the Intermediate cluster. The backbone of the CP is represented in cyan. The beads colored red and white represent the amino and carbonyl groups of the CPs in the MA(R/S)TINI 3 force field. Phosphate groups, shown in ochre, mark the lipid bilayer's boundaries. The lipid tails are rendered transparently to facilitate visualization, and the water is depicted as a continuous light-blue region.

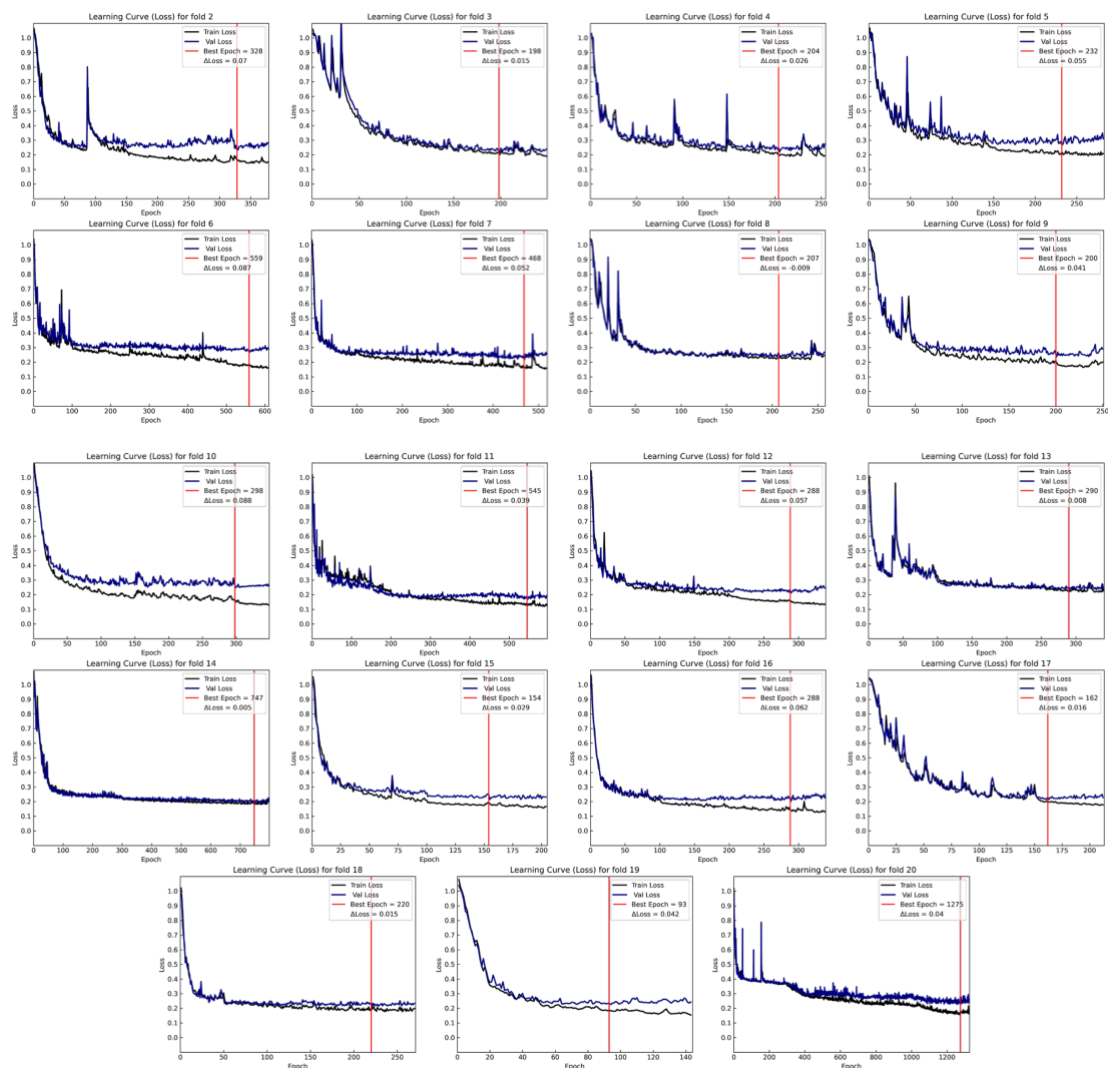


Figure S9.- Learning Curves of the different folds for the Classification task. The curves represent the decrease of the Cross Entropy loss computed for Training and Validation sets with respect to the epoch. The vertical red line represents the best epoch stored by the early stopping function.

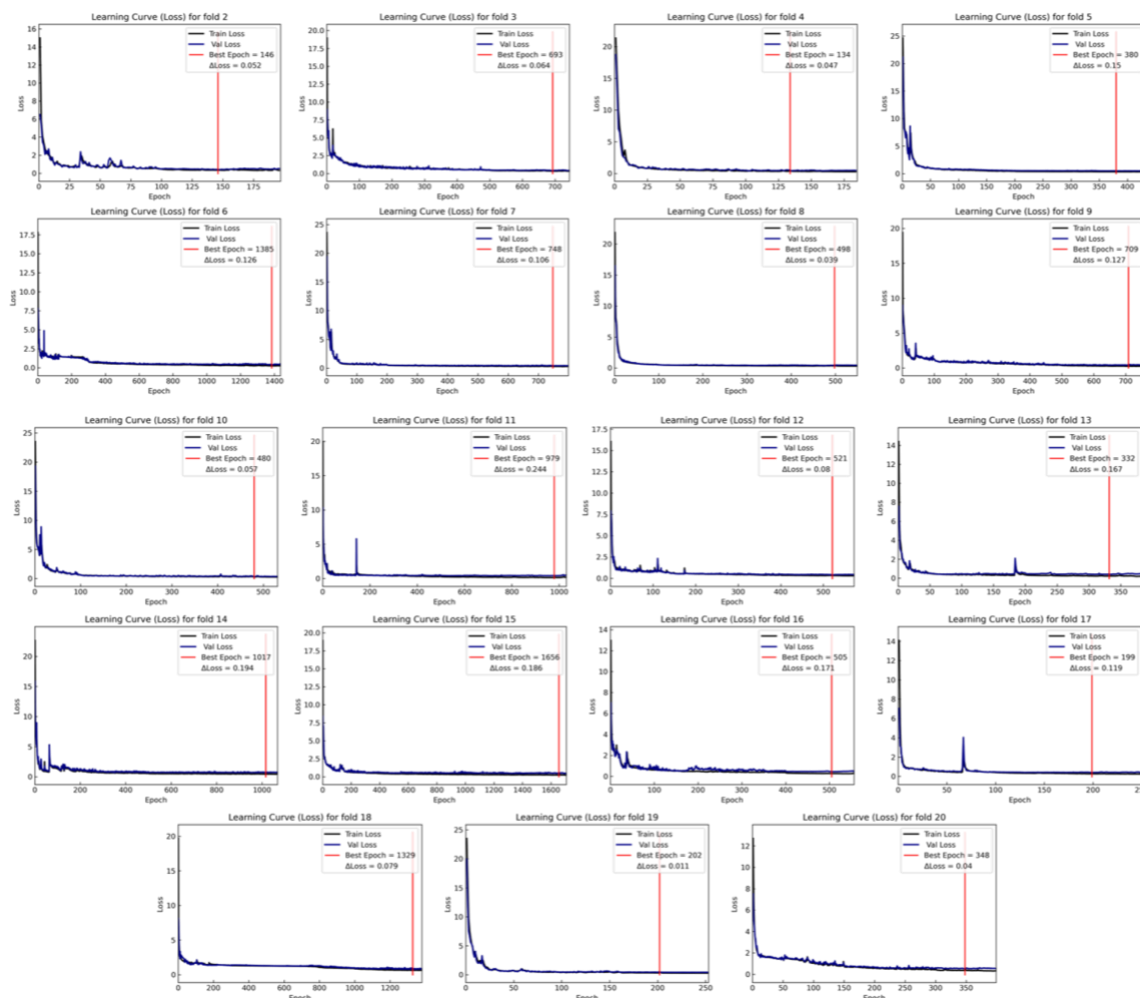


Figure S10.- Learning Curves of the different folds for the Regression task. The curves represent the decrease of the Mean Squared Error loss computed for Training and Validation sets with respect to the epoch. The vertical red line represents the best epoch stored by the early stopping function.

List of descriptor names computed from MD

The dataset comprises several categories of quantitative descriptors derived from Molecular Dynamics simulations. The system under study contains 8 peptides (referred to as CP1 through CP8), a mixed lipid bilayer composed of DMPC and DMPG, and water as the solvent.

The analysis includes **intermolecular contacts**. This category quantifies the interactions for each of the 8 peptides, as well as the entire peptide group (CP), with their surroundings. It includes contacts with other peptides (9 variables), with the general lipid pool (9 variables), and more specifically with the lipid tails (9 variables) and the polar lipid heads (9 variables). These interactions are further specified by lipid type, detailing contacts with DMPC lipids in general (9 variables), DMPG lipids (9 variables), DMPC tails (9 variables), DMPG tails (9 variables), DMPC heads (9 variables), and DMPG heads (9 variables). The solvation of the peptides is also quantified through peptide-water contacts (9 variables). Altogether this results in a total of **99** variables describing peptide intermolecular contacts.

A second group of descriptors defines the **geometric and structural properties** of the peptides. This includes the insertion depth (Z-distance) and tilt angle (Z-angle) of each of the 8 peptides relative to the membrane normal, accounting for 8 variables for each metric. The relative positioning of the peptides is captured in great detail through the center-of-mass distance, the lateral distance in the plane of the membrane, and the relative angle for

all 28 unique peptide pairs, resulting in 28 variables for each of these three geometric measurements. This results in a total of **100** variables describing geometrical and structural properties.

The third category focuses on **aggregation and local density**. The aggregation state of the peptides is monitored through 8 variables that describe the population of clusters of different sizes (from 1 to 8 peptides). Local density is assessed using a radial analysis that counts the number of molecules within 10 concentric spherical shells (from 2 Å to 20 Å). This metric is applied to peptides (10 variables), all lipid heads combined (10 variables), DMPC heads specifically (10 variables), DMPG heads (10 variables), and water molecules (10 variables), making a total of 50 radial density descriptors. This results in a total of **58** variables describing peptide aggregation and local density.

The final and most extensive category details the **non-bonded interaction energies**, separated into short-range Coulombic (Coul-SR), Lennard-Jones (LJ-SR), and Total energy terms. The interactions are systematically calculated for every component pair. First, peptide-peptide interactions are computed for all 28 unique pairs, yielding 84 energy variables in total. Next, peptide-environment interactions are assessed, covering the interaction of each of the 8 peptides with 5 distinct environmental groups (DMPC tails, DMPG tails, DMPC heads, DMPG heads, and water), which results in 120 variables. Lastly, the interactions among the environmental components themselves are calculated for all 14 unique pairs (e.g., DMPC-Tails with DMPG-Tails, DMPC-Heads with Water), contributing another 42 energy variables to the dataset. This results in a total of **246** variables describing non-bonded interaction energies.

Altogether, this comprehensive set of descriptors sums to a total of **503** unique variables for analyzing the simulation.